

Lecture 1: Introduction

MA3236 Non-Linear Programming

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Pre-requisites: linear algebra, multivariate calculus, some programming knowledge (preferably MATLAB or Python).

- Common tools: logic of necessity and sufficiency, Taylor's expansion, eigenvalues of matrices, ...
- The inherited lecture code is mostly in MATLAB. If you plan to use it, refer to https://nusit.nus.edu.sg/services/software_and_os/software/software-student/ for installation guide of MATLAB.
- See the MATLAB handout provided in Canvas if you are new to MATLAB.

Component	Weight	Format
Final exam	50%	Closed book with helpsheet
Midterm test	30%	Closed book with helpsheet
Homework 1	5%	Written assignment
Homework 2	5%	Written assignment
Coding assignment	10%	MATLAB or Python

Policy:

- Programming is required for the coding assignment, but will not be tested directly in the exams.
- You may discuss ideas with classmates, but submitted solutions and code must be your own.
- Any use of computational or AI tools should be disclosed when it affects the submitted work.

A brief history of optimization

Ancient time: geometric reasoning

- Euclid: among rectangles with fixed perimeter, the square has the largest area.
- Archimedes: the hemisphere solves an extremal volume problem for spherical caps.
- ...

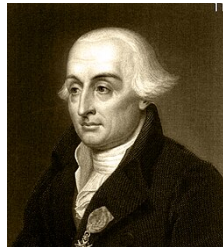
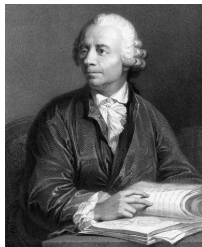
Root of “optimization”: “*optimus*” in Latin (“best” or “very good”).



A brief history of optimization (cont'd)

The 17th-18th centuries: power of calculus

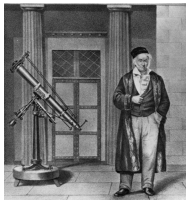
- Fermat (1636): technique to calculate maxima and minima.
- Newton (1685): numerical technique to find the roots of polynomials.
- Euler (1744), Lagrange (1755): general optimality conditions.
- ...



A brief history of optimization (cont'd)

Until 1950s: optimization before modern electronic computers

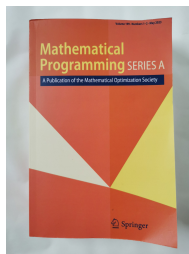
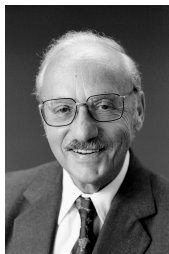
- Gauß (1801), Legendre (1805): least squares.
- Cauchy (1847): steepest descent.
- Farkas (1902): solvability of linear inequalities.
- Karush (1939), Kuhn & Tucker (1951): KKT conditions.
- ...



A brief history of optimization (cont'd)

1948-1949: birth of mathematical optimization as a modern discipline

- Koopmans (1948): “Programming in a Linear Structure” by Dantzig (1948) to “Linear Programming”.
- Dorfman (1949): “Mathematical Programming”.
- 1949: the first international conference on optimization in Chicago.



Nowadays: optimization with fast growing computing resource

Problems, algorithms, software, applications, ...

A brief history of optimization (cont'd)

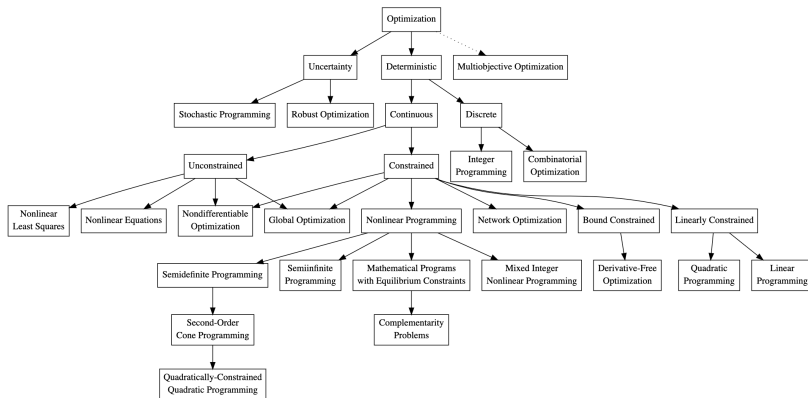


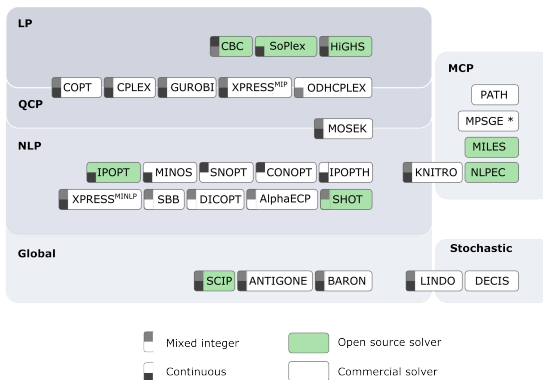
Figure: Emerging branches of optimization problems.

MA3252 - Linear and Network Optimization

MA4254 - Discrete Optimization

MA3236 - Continuous (Smooth), Nonlinear Optimization

A brief history of optimization (cont'd)



* MPSGE is not a solver, but a GAMS subsystem dedicated to solving economic equilibrium models

Figure: Overview of the solvers included with GAMS.

See also https://en.wikipedia.org/wiki/List_of_optimization_software

A brief history of optimization (cont'd)

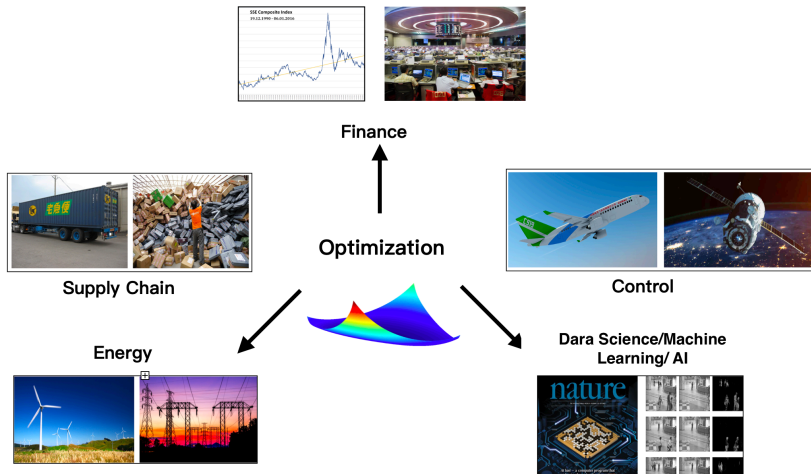


Figure: Wide applications of optimization.

Optimization model

Optimization models attempt to express the goal of solving a problem in the “best” way in mathematical terms.

Examples:

- Optimal time management
- Optimal allocation of resources
- Optimal design of manufacturing processes and instruments
- Technique in machine learning

Simple examples

Example 1. If K units of capital and L units of labor are used, a company can produce KL units of a product.

- Capital can be purchased at \$4 per unit and labor can be purchased at \$1 per unit.
- A total of \$8000 is available to purchase capital and labor.

How can the firm maximize the quantity of the product manufactured?

Simple examples

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- A total of \$8000 is available to purchase capital and labor.

How can the firm maximize the quantity of the product manufactured?

Solution. The decision variables are $K \geq 0$ and $L \geq 0$, the units of capital and labor purchased. The problem to solve is:

maximize KL

s.t. $4K + L \leq 8000, \quad K, L \geq 0.$

Example: portfolio selection

Consider an investor who has a certain amount of money to be invested in a number of different securities (stocks, bonds, etc) with random returns.

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Consider an investor who has a certain amount of money to be invested in a number of different securities (stocks, bonds, etc) with random returns.

- Let n be a positive integer. For security $i = 1, \dots, n$, let μ_i and σ_i denote its expected return and standard deviation; let ρ_{ij} denote the correlation between securities i and j .
- Let x_i be the fraction invested in security i , and define

$$\mathbf{x} := [x_1; \dots; x_n] \in \mathbf{R}^n, \quad \boldsymbol{\mu} := [\mu_1; \dots; \mu_n] \in \mathbf{R}^n,$$

$$\mathbf{Q} := (Q_{ij}) \in \mathbf{R}^{n \times n}, \quad Q_{ij} := \rho_{ij} \sigma_i \sigma_j.$$

The vector $\mathbf{x}$ is called a portfolio vector.

$$\text{expected return of } \mathbf{x} = \boldsymbol{\mu}^T \mathbf{x},$$

$$\text{variance of } \mathbf{x} = \mathbf{x}^T \mathbf{Q} \mathbf{x}.$$

- Also, $\sum_{i=1}^n x_i = 1$ and $x_i \geq 0$ for $i = 1, \dots, n$.

Example: portfolio selection (cont'd)

For a given target expected return R , a valid portfolio vector $\mathbf{x}$ is called *efficient* if it has the minimum variance among all portfolios that have at least expected return R .

Markowitz's portfolio (also called mean-variance) optimization problem:

$$\min_{\mathbf{x}} \quad \mathbf{x}^T Q \mathbf{x}$$

$$\text{s.t.} \quad \sum_{i=1}^n x_i = 1,$$

$$\boldsymbol{\mu}^T \mathbf{x} \geq R,$$

$$x_i \geq 0, \quad i = 1, \dots, n.$$

Example: sensor network localization and molecular conformation

Aim: find the positions of atoms in a molecule (typically a protein molecule) given distances estimated from **Nuclear Magnetic Resonance spectroscopy**.

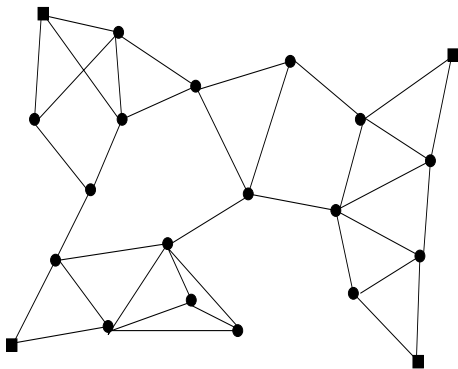


Figure: A sensor network with 4 anchors (filled squares). An edge in the graph means that the sensor-sensor or sensor-anchor pairs are within the radio range R .

Example: sensor network localization and molecular conformation (cont'd)

- Let d, m, n be positive integers; $\|\cdot\|$ denotes the Euclidean norm on $\mathbf{R}^d$.

[Unknown] Sensor positions $\mathbf{x}^1, \dots, \mathbf{x}^n \in \mathbf{R}^d$.

[Known]

- Anchor positions $\mathbf{a}^1, \dots, \mathbf{a}^m \in \mathbf{R}^d$.
- Measurement index sets $\mathcal{N} \subseteq \{1, \dots, n\}^2$ and $\mathcal{M} \subseteq \{1, \dots, m\} \times \{1, \dots, n\}$. For $(i, j) \in \mathcal{N}$, let d_{ij} be the measured sensor-sensor distance; for $(i, j) \in \mathcal{M}$, let r_{ij} be the measured anchor-sensor distance.
- Because of noise, the distances r_{ij} and d_{ij} are not estimated exactly but only approximately.
- The problem to solve is the following:

$$\min_{\mathbf{x}^1, \dots, \mathbf{x}^n \in \mathbf{R}^d} \sum_{(i,j) \in \mathcal{N}} (\|\mathbf{x}^i - \mathbf{x}^j\|^2 - d_{ij}^2)^2 + \sum_{(i,j) \in \mathcal{M}} (\|\mathbf{a}^i - \mathbf{x}^j\|^2 - r_{ij}^2)^2.$$

Example: machine learning (optional)

Given n points $\mathbf{x}^1, \dots, \mathbf{x}^n \in \mathbf{R}^d$, find the smallest ellipsoid that encloses all the points.

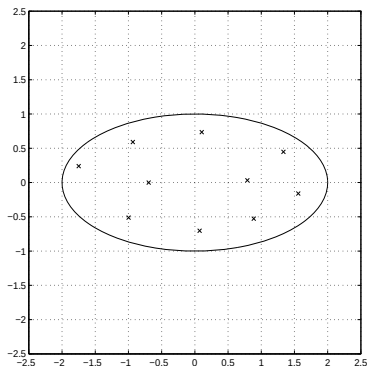


Figure: Smallest enclosing ellipsoid problem.

Application: **anomaly detection**.

Example: machine learning (optional, cont'd)

An ellipsoid with center $\mathbf{a}$ and shape matrix $\mathbf{H}$ is a set of the form

$$E(\mathbf{a}, \mathbf{H}) = \{\mathbf{x} \in \mathbf{R}^d : (\mathbf{x} - \mathbf{a})^T \mathbf{H} (\mathbf{x} - \mathbf{a}) \leq 1\},$$

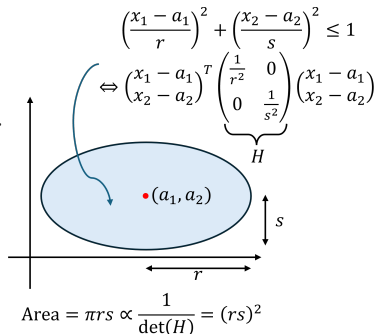
To enclose all the points, we must have

$$(\mathbf{x}^i - \mathbf{a})^T \mathbf{H} (\mathbf{x}^i - \mathbf{a}) \leq 1, \quad \forall i = 1, \dots, n.$$

The volume of the ellipsoid is proportional to $1/\det(\mathbf{H})$.

Thus we want to solve the problem:

$$\begin{aligned} \min_{\mathbf{a}, \mathbf{H}} \quad & \frac{1}{\det(\mathbf{H})} \\ \text{s.t.} \quad & (\mathbf{x}^i - \mathbf{a})^T \mathbf{H} (\mathbf{x}^i - \mathbf{a}) \leq 1, \quad \forall i = 1, \dots, n. \end{aligned}$$



General nonlinear programming (NLP) problems

$$\begin{array}{ll} \text{Minimize (or Maximize)} & f(\mathbf{x}) \\ \text{Subject to} & \mathbf{x} \in S \subseteq \mathbf{R}^n \end{array}$$

- The symbols min, max, and s. t. mean minimize, maximize, and subject to, respectively.
- The column vector $\mathbf{x}$ is the **decision variable**.
- The function $f : S \rightarrow \mathbf{R}$ which we wish to minimize (or maximize) is known as the **objective function**.
- S is known as the **feasible set**.
- A solution or a point in the feasible set is a **feasible solution** or a **feasible point**; otherwise, it is an **infeasible solution** or **infeasible point**.

Terminology and notation

$$\begin{array}{ll} \text{Minimize (or Maximize)} & f(\mathbf{x}) \\ \text{Subject to} & \mathbf{x} \in S \subseteq \mathbf{R}^n \end{array}$$

- (a) For a **minimization** problem, a feasible point $\mathbf{x}^* \in S$ is a **global minimizer** if $f(\mathbf{x}^*) \leq f(\mathbf{x})$ for every $\mathbf{x} \in S$. Define

$$\operatorname{argmin}_{\mathbf{x} \in S} f(\mathbf{x}) := \{\mathbf{x} \in S : f(\mathbf{x}) \leq f(\mathbf{y}) \text{ for every } \mathbf{y} \in S\}.$$

Thus $\mathbf{x}^* \in \operatorname{argmin}_{\mathbf{x} \in S} f(\mathbf{x})$; use equality only when the minimizer is unique.

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Thus $\mathbf{x}^* \in \operatorname{argmin}_{\mathbf{x} \in S} f(\mathbf{x})$; use equality only when the minimizer is unique.

- (b) For a **maximization** problem, reverse the inequality and define $\operatorname{argmax}_{\mathbf{x} \in S} f(\mathbf{x})$ analogously.
- The number $f(\mathbf{x}^*)$ is the **optimal value**.

Unboundedness

$$\begin{array}{ll} \text{Minimize (or Maximize)} & f(\mathbf{x}) \\ \text{Subject to} & \mathbf{x} \in S \subseteq \mathbf{R}^n \end{array}$$

- (a) A **minimization** problem is **unbounded below** if, for every $K \in \mathbf{R}$, there exists $\mathbf{x} \in S$ such that $f(\mathbf{x}) < K$. Equivalently, $\inf_{\mathbf{x} \in S} f(\mathbf{x}) = -\infty$.

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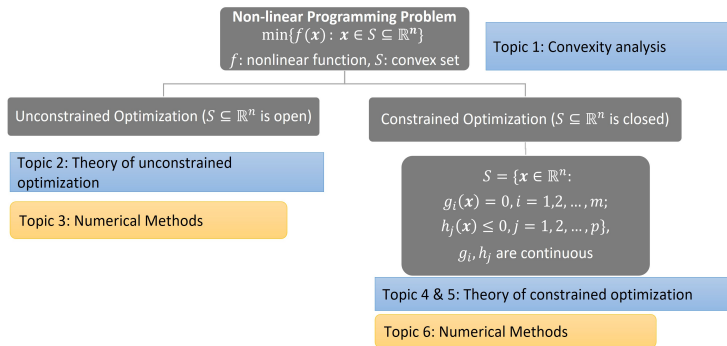
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- (b) A **maximization** problem is **unbounded above** if, for every $K \in \mathbf{R}$, there exists $\mathbf{x} \in S$ such that $f(\mathbf{x}) > K$. Equivalently, $\sup_{\mathbf{x} \in S} f(\mathbf{x}) = +\infty$.

The following optimization problems are **equivalent**:

$$\begin{array}{ll} \text{Maximize} & f(\mathbf{x}) \\ \text{Subject to} & \mathbf{x} \in S \end{array} \Leftrightarrow \begin{array}{ll} \text{Minimize} & -f(\mathbf{x}) \\ \text{Subject to} & \mathbf{x} \in S \end{array}$$

- $\mathbf{x}^* \in S$ is an optimal solution for the maximization problem with optimal objective value $f(\mathbf{x}^*)$ if and only if it is also an optimal solution for the minimization problem with optimal objective value $-f(\mathbf{x}^*)$.
- Else, both problems are infeasible or both problems have unbounded objective value.
- We will mostly focus on the discussion of minimization problems.

Topics



Tentative schedule:

- Topic 1: Lectures 3-5 (Weeks 2-3).
- Topic 2: Lectures 6-7 (Week 4).
- Topic 3: Lectures 8-11 (Weeks 5-6).
- Online consultation: Thurs. 1 October (Week 7).
- Midterm: Mon. 5 October (Week 8).
- Topics 4 & 5: Lectures 12-18 (Weeks 7-11).
- Topic 6: Lectures 19-22 (Weeks 11-13).
- Final: to be announced.

An unconstrained nonlinear programme

Let n be a positive integer, let $\mathcal{X} \subseteq \mathbf{R}^n$ be open, and let $f : \mathcal{X} \rightarrow \mathbf{R}$.

Minimize (or Maximize) $f(\mathbf{x})$
 $\mathbf{x}$

Subject to $\mathbf{x} \in \mathcal{X}$

- **Objective function** $f : \mathcal{X} \rightarrow \mathbf{R}$ is a nonlinear function of $\mathbf{x}$.
- **Feasible set** $\mathcal{X}$ is an **open** subset of $\mathbf{R}^n$; in particular, it may be $\mathbf{R}^n$.

Definition 1.1 (Open set)

For $\mathbf{x} \in \mathbf{R}^n$ and $\epsilon > 0$, define the open ball

$$B(\mathbf{x}, \epsilon) := \{\mathbf{y} \in \mathbf{R}^n : \|\mathbf{y} - \mathbf{x}\| < \epsilon\}.$$

A set $S \subseteq \mathbf{R}^n$ is **open** if every $\mathbf{x} \in S$ has some $\epsilon > 0$ for which $B(\mathbf{x}, \epsilon) \subseteq S$.

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Example of an unconstrained NLP: the molecular conformation problem

$$\min_{\mathbf{x}^1, \dots, \mathbf{x}^n \in \mathbf{R}^d} \sum_{(i,j) \in \mathcal{N}} (\|\mathbf{x}^i - \mathbf{x}^j\|^2 - d_{ij}^2)^2 + \sum_{(i,j) \in \mathcal{M}} (\|\mathbf{a}^i - \mathbf{x}^j\|^2 - r_{ij}^2)^2.$$

A constrained nonlinear programme

For $i = 1, \dots, m$ and $j = 1, \dots, p$, let $g_i : \mathbf{R}^n \rightarrow \mathbf{R}$ and $h_j : \mathbf{R}^n \rightarrow \mathbf{R}$, respectively.

$$\begin{array}{ll}\text{Minimize (or Maximize)} & f(\mathbf{x}) \\ \text{Subject to} & \mathbf{x} \in \mathbf{R}^n \\ & g_i(\mathbf{x}) = 0, \quad i = 1, 2, 3, \dots, m, \\ & h_j(\mathbf{x}) \leq 0, \quad j = 1, 2, 3, \dots, p,\end{array}$$

- f is the **objective function**; the g_i 's and h_j 's are the equality- and inequality-constraint functions.
- Some of these functions may be nonlinear. If all g_i and h_j are continuous, the **feasible set**

$$S := \{\mathbf{x} \in \mathbf{R}^n : g_i(\mathbf{x}) = 0, \quad i = 1, \dots, m; h_j(\mathbf{x}) \leq 0, \quad j = 1, \dots, p\}$$

is a **closed** subset of $\mathbf{R}^n$.

A constrained nonlinear programme

Examples of constrained NLP:

① Portfolio selection model

$$\begin{aligned} \min_{\mathbf{x}} \quad & \mathbf{x}^T \mathbf{Q} \mathbf{x} \\ \text{such that} \quad & \sum_{i=1}^n x_i = 1, \\ & \boldsymbol{\mu}^T \mathbf{x} \geq R, \\ & x_i \geq 0, \quad i = 1, \dots, n. \end{aligned}$$

② Finding the smallest ellipsoid that encloses all the given points

$$\begin{aligned} \min_{\mathbf{a}, \mathbf{H}} \quad & \frac{1}{\det(\mathbf{H})} \\ \text{s.t.} \quad & (\mathbf{x}^i - \mathbf{a})^T \mathbf{H} (\mathbf{x}^i - \mathbf{a}) \leq 1, \quad \forall i = 1, \dots, n. \end{aligned}$$

Closed set

Remark. A set $S \subseteq \mathbf{R}^n$ is closed if its complement is open; equivalently, S is open if its complement is closed.

Proposition 1.2 (Equivalent characterization of a closed set)

A set $S \subseteq \mathbf{R}^n$ is closed if and only if, whenever $\mathbf{x}_k \in S$ and $\mathbf{x}_k \rightarrow \mathbf{x}$, the limit $\mathbf{x}$ also belongs to S . The empty set and $\mathbf{R}^n$ are both open and closed.

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Example 1.3

(a) Let $a, b \in \mathbf{R}$ with $a < b$. The interval $[a, b]$ is closed in $\mathbf{R}$.

The interval $(a, b]$ is not closed in $\mathbf{R}$.

Indeed, $x_k := a + \frac{b-a}{k} \in (a, b]$, but $x_k \rightarrow a \notin (a, b]$.

(b) Every box $\{\mathbf{x} \in \mathbf{R}^n : a_i \leq x_i \leq b_i, i = 1, \dots, n\}$ is closed.

Question. Let q be a positive integer, and let $S_1, \dots, S_q \subseteq \mathbf{R}^n$ be closed. Is $\bigcap_{i=1}^q S_i$ closed?

Closed set

The following result is useful to show if a set is closed.

Proposition 1.4 (Continuous constraints define closed sets)

Let $g : \mathbf{R}^n \rightarrow \mathbf{R}$ be a **continuous** function. Then

- (a) The set $\{\mathbf{x} \in \mathbf{R}^n : g(\mathbf{x}) \leq 0\}$ is closed.
- (b) The set $\{\mathbf{x} \in \mathbf{R}^n : g(\mathbf{x}) \geq 0\}$ is closed.
- (c) The set $\{\mathbf{x} \in \mathbf{R}^n : g(\mathbf{x}) = 0\}$ is closed.

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- (c) The set $\{\mathbf{x} \in \mathbf{R}^n : g(\mathbf{x}) = 0\}$ is closed.

Example 1.5

Verify that the set $C := \{(x, x^2) : x \in \mathbf{R}\}$ is closed.

Solution. Define $g : \mathbf{R}^2 \rightarrow \mathbf{R}$ by $g(\mathbf{x}) := x_2 - x_1^2$. Then

$$C = \{\mathbf{x} \in \mathbf{R}^2 : g(\mathbf{x}) = 0\}.$$

Since g is continuous, C is closed by Proposition 1.4 (c).

Closed set

Corollary 1.6 (Closed feasible set)

*Suppose $g_i, h_j : \mathbf{R}^n \rightarrow \mathbf{R}$ are continuous for $i = 1, \dots, m$ and $j = 1, \dots, p$.
The set*

$$S := \{\mathbf{x} \in \mathbf{R}^n : g_i(\mathbf{x}) = 0, \ i = 1, \dots, m; h_j(\mathbf{x}) \leq 0, \ j = 1, \dots, p\}$$

is closed.

Note that S is the feasible set of the constrained NLP.

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Example 1.7

Define the continuous functions $h_1(\mathbf{x}) := x_1^2 + x_2^2 - 3$ and $h_2(\mathbf{x}) := -x_1 + 2 \sin x_2$. Then

$$S := \{\mathbf{x} \in \mathbf{R}^2 : h_1(\mathbf{x}) \leq 0, h_2(\mathbf{x}) \leq 0\}$$

is closed. Equivalently, $S = \{\mathbf{x} \in \mathbf{R}^2 : x_1^2 + x_2^2 \leq 3, x_1 - 2 \sin x_2 \geq 0\}$.

Take-home messages

- Optimization is not new and has been ever growing.
- Optimization problem is prevalent; it describes mathematically what you want (f), what you have (S), and what you want to change ($\mathbf{x}$).

$$\min \text{ (or max) } f(\mathbf{x}), \quad \text{s. t. } \mathbf{x} \in S.$$

- Before you embark on solving an optimization problem, think about its well-posedness.